Laith Hijazi

Halifax, NS | Phone: (902)-489-0536| **Email:** laith.hijazi.1999@gmail.com | Portfolio

Education

Dalhousie University — Bachelor of Science in Biochemistry and Molecular Biology
Graduated: May 2021

Nova Scotia Community College — IT Web Programming
Graduated: May 2024

Skills: Python (NumPy/Pandas/PyTorch/Keras/scikit-learn), PyQt5/Qt, JavaScript/TypeScript, React, C#, ASP.NET, Java, PHP, Laravel, SQL (MySQL/PostgreSQL), MongoDB, Arduino/Embedded Systems, Docker, Git/Version Control, Machine Learning/AI, Data Pipeline Engineering, Hardware-Software Integration, Serial Communication Protocols, Real-time Systems, Scientific Data Analysis, Mass Spectrometry Software, Bash/Linux

Work Experience

- ***Scientific Software Developer | Data Scientist*** ***Greenlight Analytical*** *April 2024 - Present*
  - Engineered end-to-end software solutions for mass spectrometry instrumentation, developing PyQt5-based desktop applications integrating real-time hardware control, data acquisition pipelines, and advanced analytics for chemical analysis in agriculture and laboratory sciences.
  - Designed and implemented machine learning models (Random Forest, Neural Networks) for classification and regression of cannabis phytochemicals (terpenes and cannabinoids), achieving high accuracy in compound identification and quantification from mass spectrometry data.
  - Developed hardware-software interfaces for Arduino microcontroller systems and Extorr RGA firmware, implementing serial communication protocols and real-time state management for precise control of vacuum systems, thermal regulators, and analytical instruments.
  - Built scalable data processing pipelines for spectral data analysis, including automated calibration workflows, baseline correction algorithms, peak detection systems, and database management for experimental results and instrument metadata.
  - Collaborated cross-functionally with scientists and engineers to translate complex analytical chemistry requirements into intuitive user interfaces and robust backend systems, optimizing instrument performance and data quality for research applications.
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- ***Data Annotation*** ***LifeRaft*** *Summer 2023 Work Term*
  - Executed meticulous data annotation on social media posts, categorizing them by specific threat types, contributing to the development of a new feature for threat intelligence solutions at an OSINT software company.
  - Spearheaded the incorporation of Language Models (LLMs) into threat intelligence reports, enhancing the summarization of related post clusters, resulting in more efficient and insightful reporting processes.

- ***STEM Tutor*** ***Paper Education*** *April 2021 – September 2023*
  - Provided tutoring services in biology, chemistry, math, and computer science via the Socratic method to over 2000+ students since starting.
  - Conducted multiple simultaneous text and audio based tutoring sessions while judiciously using an in-built whiteboard as well as free online educational resources.
  - Adapted teaching approach to student age and level of knowledge.
  - Provided mutual support to other tutors when difficult academic questions arose.